

CoreOps Week 2 Mon Module

Kubernetes Internals & High Availability Architecture

Objective: Master control plane internals, node lifecycle, upgrade patterns, HA design, static pod behaviour, and failure-handling logic.

MONDAY: kube-apiserver, scheduler, controller-manager

Core Concepts

- kube-apiserver internal request flow
 - Authentication → Authorization → Admission Control → Storage
- How the API server talks to etcd
- Scheduler internals
 - Scheduling queue
 - Filtering & scoring plugins
 - Leader election mechanics
- Controller-manager
 - Node controller
 - ReplicaSet controller
 - EndpointSlice controller

Key Outcomes

You understand the **entire lifecycle of a kubectl apply**

You understand **why pods go Pending, why API server spikes, and how controllers converge state**

WHAT TO PRACTICE (HANDS-ON)

All exercises run on your Kubeadm cluster (you will create this cluster in the assignment).

Lab 1: Observe API server request flow

```
kubect proxy --port=8080  
curl localhost:8080/api
```

Check authentication: `kubect auth can-i get pods`

Lab 2: Break & observe scheduler behaviour

Stop scheduler static pod: `sudo mv /etc/kubernetes/manifests/kube-scheduler.yaml /root/`

Observe: `kubect get pods -A`

`kubect describe pod <pod-name>`

Lab 3: Controller-manager failure test

`sudo mv /etc/kubernetes/manifests/kube-controller-manager.yaml /root/`

Observe pod resync behaviour.

THE DEVOPS WAR ROOM

WHERE TO STUDY

- Kubernetes Docs: API Server:
<https://kubernetes.io/docs/concepts/overview/components/>
 - Scheduler Internals (High quality):
<https://github.com/kubernetes/community/tree/master/contributors/devel/sig-scheduling>
 - KCM Deep Dive (KubeCon): Search “Controller Manager KubeCon Deep Dive”
-

INFRATHRONE MINDSET NOTES

- The API server is your single point of truth. Treat it like a database.
 - Scheduler is not a pod mover; it decides, and kubelet executes.
 - Controllers fix drift. They do not create pods randomly.
 - If the control plane is slow, your whole company is slow.
-

